

From Concept to Submission in 8 Hours

An Agentic Research Workflow

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Building the Infrastructure

- **OpenClaw setup:** Configured an AI agent as my research assistant, connected via Telegram
- **Local models (LM Studio):** Tool calling was unreliable — had to pivot
- **DeepSeek API:** The breakthrough — cheap (~\$0.15/M input), fast, reliable
- **Qwen via Ollama:** Second model for cross-checking (different architecture)
- **Whisper:** Speech-to-text for dictating ideas while pacing the kitchen
- **Dependencies:** R + glmnet/furrr, Python + matplotlib/whisper, Pandoc + LaTeX

Lesson: Start with the API that works, optimize later.

PROCOVA's Blind Spot

- **PROCOVA** is EMA-qualified for improving trial efficiency via prognostic scores
- But it assumes the **external score is perfectly calibrated** for the trial population
- **Population shift** is the rule, not the exception — especially in oncology
- **No existing method** diagnoses or corrects miscalibration using blinded trial data

Ridge-Cal

Treat the external score as a **pre-trained model** and apply a **regularized correction** on blinded data:

- **Diagnose:** Compare C-index of score alone vs. score + calibration covariates
- **Calibrate:** Ridge-penalized Cox regression on blinded data, λ by CV
- **Analyze:** Standard Cox PH with calibrated score + robust sandwich SE

6 parameters, blinded data, no additional data collection needed.

May 17–18, 2026

Time	Milestone
~21:00	Concept brainstorm (TMLE + Bayesian MCMC — too complex, scrapped)
~22:00	Pivot to ridge regression on 5 covariates
~23:00	First simulation — agent found a bug (non-PH assumption)
~00:00	10K-rep simulation running in background
~01:00	MAP-Cox bug surfaced (k applied after pooling)
~02:00	Bug fixed, re-simulation triggered

Time	Milestone
~06:00	Manuscript drafted — agent wrote prose, I reviewed every section
~07:00	Sent to Gemini — Major Revision, 4 issues
~08:00	Sent to Qwen for independent review — 7 MORE issues
~09:00	Digital twin landscape report drafted
~10:00	JBS formatting, self-verify, corrections
~14:00	Voice note → small strata investigation
~15:00	Workflow codified into reusable skill

What Came Out of It

Deliverable	Format	Audience
Ridge-Cal manuscript	PDF, 12 pages	JBS submission
Response to Reviewer 2	Markdown	Journal
Digital twin survey	PDF, 12 pages	BLT (BARDS Leadership T
Small strata white paper	Markdown	Internal stat leads
Research Workflow skill	SKILL.md	Future projects
Simulation code	GitHub (public)	Reproducibility

Why Not One AI?

Same-model reviewers share the same **blind spots**.

Internal Loop (my agent, ~\$0.15/M tokens)

- Primary: DeepSeek v4 Flash | Reviewer: Qwen
- Caught: Section 4 redundancy, δ justification, missing data, table headers

External Loop (separate company)

- Gemini Pro — completely different training
- Caught: Non-collapsibility, tone, event rates, LoRA framing

Review Results

Round	Reviewer	Verdict	Issues
1	Gemini	Major Revision	4
2	Gemini	Accept	—
3	Qwen	Major Revision	7 (different)
4	Qwen	Minor Revision	
5	Qwen	Accept	

Qwen caught things Gemini missed: redundancy, δ justification, table headers, MAP-Cox framing, sandwich variance scope, missing data, references.

Takeaway: Model-diverse review converges faster than any single reviewer.

10 Minutes Start to Finish

Input: Voice note while pacing the kitchen (Whisper transcription)
“Do we need to pool small strata for CMH and MN methods?”

Step	Time	How
Transcription	< 1 min	Whisper
Problem scoping	1 min	I dictated, agent structured
Independent review	1 min	Qwen caught framing weakness
Revision	1 min	I directed, agent refined
Code + simulation	3 min	Agent drafted code, 5K reps
Code review v1	1 min	Qwen caught 3 bugs
Code review v2	1 min	Verified fixes
Final + push	1 min	White paper to GitHub

Three Rounds

Review	Verdict	What It Caught
v1	Minor Issues	Scenario 4 duplicated, CMH RR variance unstratified, Wald mislabeled as MN
v2	Minor Issues	Scenario 4 label misleading, MN SE approximation
Final	Pass	Verified against bootstrap empirical variance

All bugs fixed before delivery.

Research Skill File

Phase	Description
Ph 0: Topic ID	Identify gap, map contradictory literature
Ph 1: Writeup	Write, spawn isolated reviewer, iterate
Ph 2: Simulate	2 $\rightarrow$ 20 $\rightarrow$ 200 $\rightarrow$ 10K reps
Ph 3: QC	Code review before big runs — mandatory
Ph 4: Full Run	Background, scheduled, monitored
Ph 4.5: Audit	Verify code matches manuscript claims
Ph 5: Manuscript	Write, verify refs, PDF self-check
Ph 6: Revision	Parse $\rightarrow$ Reconcile $\rightarrow$ Re-sim
Ph 7: Submit	Reviewer $\rightarrow$ Revise $\rightarrow$ Max 3

Key Lessons Codified

- ① **Isolate reviewers** — fresh sessions, no cross-contamination
- ② **Diversify models** — different AIs catch different bugs
- ③ **Code review BEFORE big simulations** — trust but verify
- ④ **PDF self-verify** — automated checks before reaching humans
- ⑤ **Clean up cruft** — stale files cause confusion
- ⑥ **Push to GitHub at milestones** — enables external review
- ⑦ **Batch deliveries** — pace yourself

Results

- **Independent AI reviewers caught 11 bugs** before human eyes saw the draft
- **Small tests before big runs** — avoided ~5 dead ends
- **Human-in-the-loop essential** — I directed 8 reframing cycles
- **The agent amplified my judgment, did not replace it**

Lessons Learned

- **PDF generation was painful** — 5 failed attempts with silent image failures. Now has a verification script.
- **Code review almost skipped** — hard checkpoint now in the pipeline.
- **Stale artifacts accumulated** — explicit cleanup at every milestone now.

Economics

- **Session cost:** ~500K DeepSeek tokens, **well under \$1.00**
- **That bought:** 1 manuscript, 1 BLT report, 1 white paper, 1 reusable skill, 3 AI reviews from 2 models, 3 GitHub repos
- **Key insight:** The agent replaces the drafting lag, not the statistician

Thank You

Repos: - github.com/doublerobust/ridge-cal -
github.com/doublerobust/small-strata-pooling

The skill: SKILL.md in my OpenClaw workspace — available on request.